Archit Uniyal

Email: a.uniyal@virginia.edu
Phone: (+1) 4348330881
GitHub
LinkedIn
Website

Skills

Python, C++, C, SQL, Java, Javascript, research, Differential Privacy, Privacy-preserving ML, Natural Language Processing, Computer Vision, Pruning Algorithms, Pytorch, Tensorflow, Bash, Linux, Git, AWS, Google Cloud, Apache arrow, Postgres, VSCode, Rstudio, HPC clusters, Hugging Face, OpenAI API, LangChain

Education

University of Virginia, Charlottesville, VA
Masters in Computer Science — August 2022 – December 2024
CGPA: 4.0

University Institute of Engineering and Technology, Panjab University, Chandigarh
B.E in Computer Science — August 2018 – July 2022

Publications

An Empirical Analysis of Memorization in Fine-tuned Autoregressive Language Models
*Mireshghallah, F., **Uniyal, A.**, Wang, T., Evans, D. K., Berg-Kirkpatrick, T.*
Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022

Quantifying Privacy Risks of Masked Language Models Using Membership Inference Attacks
*Mireshghallah, F., Goyal, K., **Uniyal, A.**, Berg-Kirkpatrick, T., Shokri, R.*
Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022

DP-SGD vs PATE: Which Has Less Disparate Impact on Model Accuracy?
***Uniyal, A.**, Naidu, R., Kotti, S., Singh, S., Kenfack, P. J., Mireshghallah, F., Trask, A.*
ML4Data Workshop at ICML 2021

GC-NET for classification of glaucoma in the retinal fundus image
*Juneja, M., Thakur, N., Thakur, S., **Uniyal, A.**, Wani, A., Jindal, P.*
Machine Vision and Applications 31, 38 (2020)

DC-Gnet for detection of glaucoma in retinal fundus imaging
*Juneja, M., Thakur, S., Wani, A., **Uniyal, A.**, Thakur, N., Jindal, P.*
Machine Vision and Applications 31, 34 (2020)

Work Experience

Software Developer Intern at OmniVision, Santa Clara, CA, USA — *May 2024 – Aug 2024*

Researched state-of-the-art automated unit testing tools and developed a Python-based automated unit testing tool using machine learning algorithms. This project reduced multiple package dependencies and facilitated the transition from search-based to machine learning-based unit test generation.

Graduate Research Assistant, University of Virginia — *Aug 2022 – May 2024*
Conducted research on the link between fine-tuning data distribution and compression algorithms in property inference attacks. Explored unlearning effects in LLMs and various auditing methods. Identified that tweaking the LLM head poses the highest privacy risk among fine-tuning approaches.

Research Scientist Intern at Oracle, Burlington, MA, USA — *May 2023 - August 2023*
Conducted research on privacy in large language models (LLMs) as well as introduced a new concept of **entity-relationship privacy** in LLMs and registered it for a patent at Oracle

Research Intern at Samsung Research Institute, Noida, India — *January 2022 – July 2022*
Conducted research on improving the existing time series-based human activity recognition algorithms in wearable trackers and on GAN-based models for improving quality of videos in samsung smartphones

Researcher at Openmined — *November 2020 – July 2022*
Conducted research on the disparate impact of DP-SGD and PATE on minority groups in the dataset.

Research Intern at Design Innovation Centre, Panjab University — *June 2019 – July 2019*
Proposed a custom architecture for image classification and an algorithm to calculate DDLS and ISNT of a fundus image using image segmentation.

Projects

An Empirical Analysis of Memorization in Fine-tuned Autoregressive Language Models
This project focuses on the memorization in large language models due to fine-tuning. We analyze different fine-tuning methods and observe that fine-tuning the head of the models makes it most vulnerable to attacks.

FireBuster
Developed a wildfire forecasting model using time series analysis and regression. Utilized PostgreSQL for data storage and psycopg2 for data access. Conducted data analysis to identify high-correlation features, visualized with Plotly and Seaborn. Trained a random forest model and evaluated performance using mean absolute error.

Indian Sign Language
Developed an interface to detect Indian Sign Language alphabets and convert them to text using InceptionResnetV3. Implemented LSTMs to generate text sequences and integrated Google Text-to-Speech API for audio output.